

Magnetic Earth Models – Background Information and Additional Resources

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Background information:

- The Earth has a north and a south magnetic pole as if there were a **giant bar magnet** in the centre of the planet (Olsen & Pauluhn, 2019) (see **Figure 1**).
- The magnetic poles have opposite positive and negative polarities. Invisible magnetic field lines travel from the north magnetic pole to the south magnetic pole as closed loops (NASA, 2021) (see **Figure 1**).
- The magnetic field extends out from Earth into space and forms the **magnetosphere** (Olsen & Pauluhn, 2019) (see **Figure 2**).
- The magnetosphere acts as a huge protective 'force field' and shields the planet from harmful particles emitted by the Sun (Olsen & Pauluhn, 2019) (see **Figure 2**).
- Most of the magnetic field is generated by the movement of liquid iron in the Earth's outer core (see **Figure 3**). This process is known as the **geodynamo** (NASA, 2021).
- The Earth has two different sets of **poles** (Dobrijevic, 2022):
 1. The **geographic poles** are fixed locations where the Earth's lines of longitude meet. The geographic south pole is in Antarctica, and the geographic north pole is in the Arctic Ocean.
 2. The **magnetic poles** are where the magnetic field lines enter Earth. These are not in the same positions as the geographic poles, and their location changes over time.
- The Earth's geographic north pole is currently a magnetic south pole (see **Figure 1**). This is why a compass needle points north – it is attracted to the magnetic south pole in the northern hemisphere (Dobrijevic, 2022).

- Earth's magnetic field is strongest at the **poles**, where the field lines converge and are vertical. At the **equator**, magnetic field lines are horizontal, and field strength is the weakest (Dobrijevic, 2022).

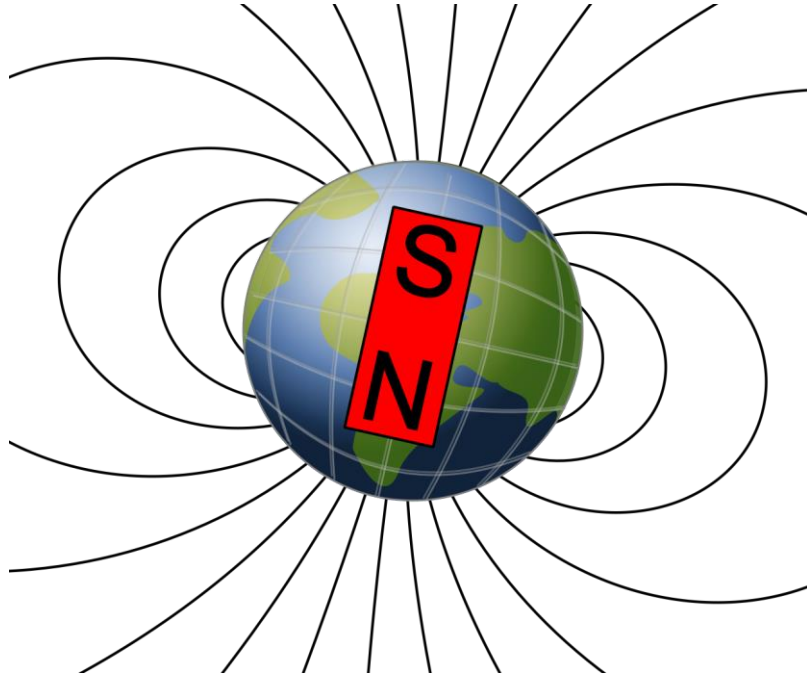


Figure 1: Earth's magnetic field lines form closed loops around the planet. Magnetic north is in the southern hemisphere, and magnetic south is in the northern hemisphere.
(Zureks, [CC0 1.0](#), via [Wikimedia Commons](#))

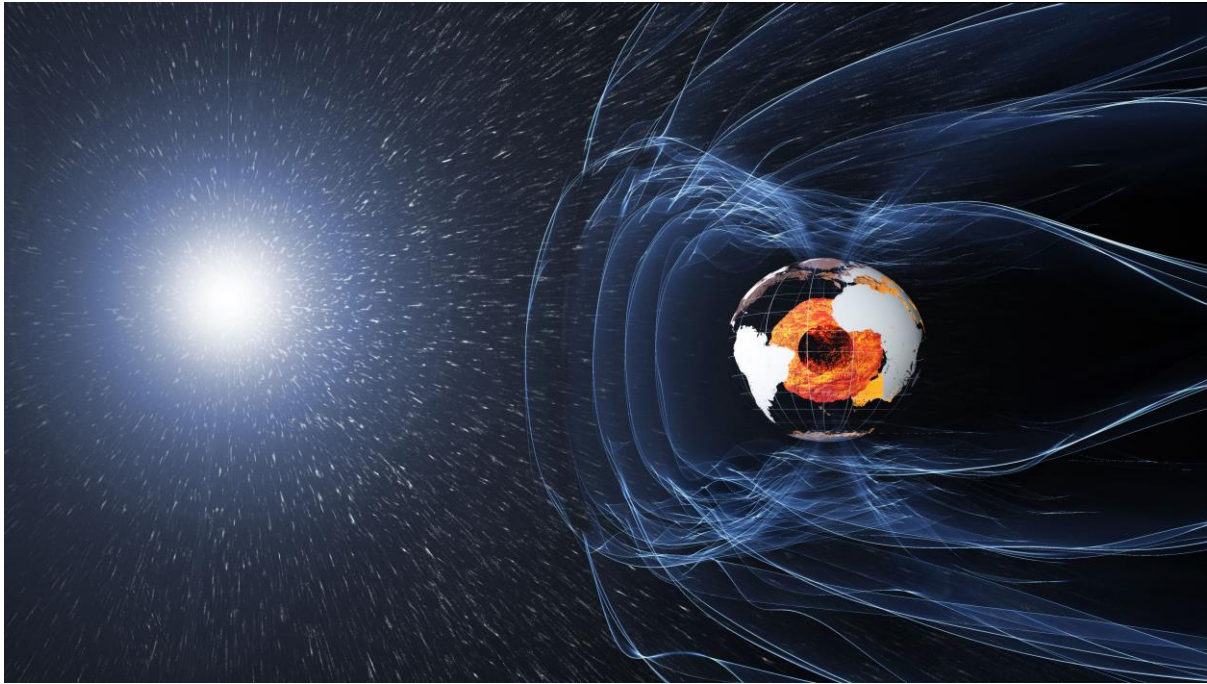


Figure 2: The Earth's magnetosphere acts as a protective force field and shields our planet from harmful solar particles. ([ESA/ATG medialab](#))

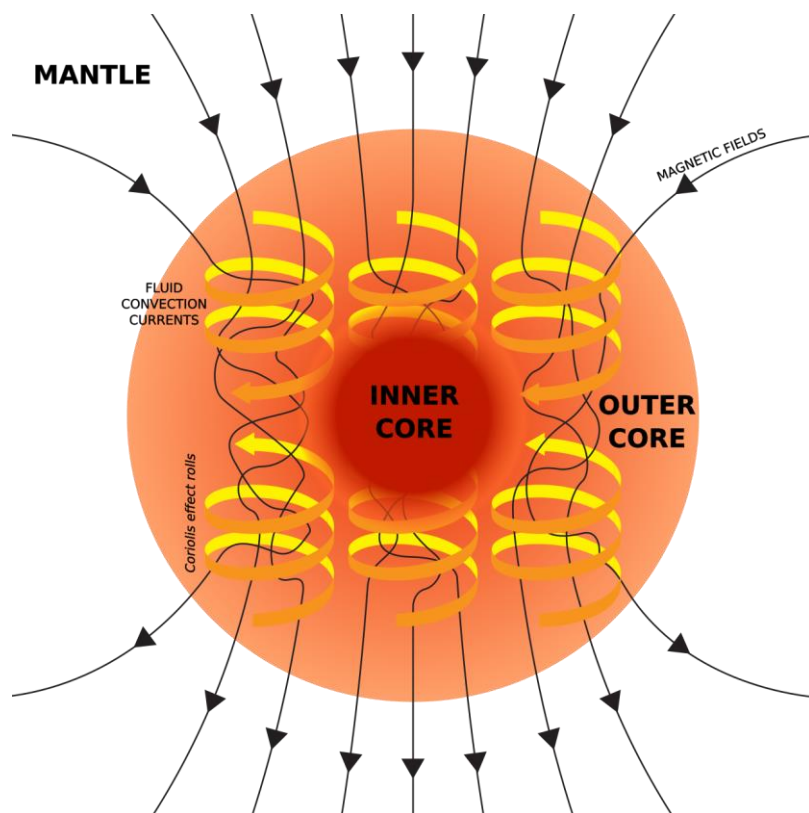


Figure 3: Earth's magnetic field is generated by the movement of liquid iron in the outer core. (Andrew Z. Colvin, [CC BY-SA 4.0](#), via [Wikimedia Commons](#))

References:

Dobrijevic, D. (2022). *Earth's magnetic field: Explained*. Available at: <https://www.space.com/earths-magnetic-field-explained>

NASA (2021). *Earth's Magnetosphere: Protecting Our Planet from Harmful Space Energy*. Available at: <https://science.nasa.gov/science-research/earth-science/earths-magnetosphere-protecting-our-planet-from-harmful-space-energy/>

Olsen, N. and Pauluhn, A. (2019). Exploring Earth's magnetic field – Three make a Swarm. *Spatium*, 2019(43), pp. 3-15. Available at: https://backend.orbit.dtu.dk/ws/portalfiles/portal/185293289/Exploring_Earths_magnetic_field.pdf

Additional resources:

For other classroom activities to introduce students to the basic concepts of magnetism: Stanford University (2015-2020). *A Teacher's Guide to Magnetism and Solar Activity*. Available at: <https://solar-center.stanford.edu/activities/Magnetism-Solar-Activity/>

For further information on the generation of the Earth's magnetic field and how it is studied: Olsen, N. and Pauluhn, A. (2019). Exploring Earth's magnetic field – Three make a Swarm. *Spatium*, 2019(43), pp. 3-15. Available at: https://backend.orbit.dtu.dk/ws/portalfiles/portal/185293289/Exploring_Earths_magnetic_field.pdf

A short ESA animation showing Earth's magnetic field acting as a protective shield around the planet: European Space Agency (ESA) (2012). *The force that protects our planet* [Video]. ESA Multimedia. Available at: https://www.esa.int/ESA_Multimedia/Videos/2012/09/The_force_that_protects_our_planet